SCHOOL MANAGEMENT SYSTEM FOR MATUNDU PRIMARY SCHOOL

FINAL PROJECT REPORT

Mwange Musa Muthami

Registration Number: 23/05037

Submission Date: April 2026

Supervisor: Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu

A project document submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Information Technology (BIT 3105)

ABSTRACT

This report presents the design, development, and deployment of a web-based School Management System for Matundu Primary School, Kitui County. The system automates student management, attendance tracking, CBC-aligned assessment and grading, report card generation, fee management, timetabling, and communication. Built with React 18, Node.js/Express, and PostgreSQL, the system was deployed on Cloudflare Pages and Render. All 11 modules were successfully implemented, tested, and deployed, achieving 100% functional requirement coverage.

CHAPTER 1: INTRODUCTION

(See Proposal Document for complete Background, Problem Statement, Objectives, and Significance sections.)

CHAPTER 2: LITERATURE REVIEW

(See Proposal Document for complete Literature Review.)

CHAPTER 3: METHODOLOGY

(See Proposal Document for complete Research and Development Methodology.)

CHAPTER 4: SYSTEM DESIGN

(See Design Specification Document for complete System Architecture, Database Design, Module Design, Frontend Design, and Security Design.)

CHAPTER 5: IMPLEMENTATION

5.1 System Completion Status

Module	Status	Completion
Authentication & RBAC	Complete	100%
Student Management	Complete	100%
Staff Management	Complete	100%
Academic Structure (CBC)	Complete	100%
Attendance Tracking	Complete	100%
Assessment & Grading	Complete	100%
Report Cards	Complete	100%
Fees Management	Complete	100%
Communication	Complete	100%
Timetable	Complete	100%
Dashboard (Admin + Student)	Complete	100%
Deployment	Complete	100%

5.2 Database Statistics

23+ database tables, 6 grades, 7 classes, 10 teachers, 70+ students, full assessment data with scores, fee structures with invoices and payments, auto-generated timetable.

5.3 Key Technical Achievements

- Fully automated CBC competency grading engine
- Role-based dashboards with real-time analytics
- Audit logging for all critical system operations
- Redis-backed rate limiting for API security
- Auto-repair mechanism for timetable and assessment data on server startup

CHAPTER 6: TESTING

(See Test Plan Document for complete test strategy, test cases, and results.)

Summary: 46 test cases executed across all modules. 100% pass rate achieved.

CHAPTER 7: CONCLUSION AND RECOMMENDATIONS

7.1 Conclusion

The School Management System for Matundu Primary School has been successfully designed, developed, tested, and deployed. The system addresses all identified problems: manual record-keeping, data loss risk, slow report generation, poor parent communication, and lack of real-time analytics. All 11 core modules are fully functional and accessible to stakeholders.

7.2 Recommendations for Future Work

- M-Pesa STK Push Integration: Direct mobile money payment initiation from within the system.
- SMS/Email Notifications: Automated alerts for attendance, fees, and announcements.
- Flutter Parent App: Native mobile application for parents to access student data.
- PDF Report Card Generation: Server-side PDF generation for downloadable report cards.
- Multi-School Support: Architecture extension to support multiple schools under one platform.

REFERENCES

Ministry of Education. (2023). National Guidelines on Basic Education Information Management Systems. Republic of Kenya.

Pressman, R. S. (2014). Software Engineering: A Practitioner's Approach (8th ed.). McGraw-Hill.

ShulePro. (2021). School Administration Software Overview. Retrieved from https://www.shulepro.com

Sommerville, I. (2015). Software Engineering (10th ed.). Pearson.

Zeraki Analytics. (2023). Transforming Education through Data. Retrieved from https://zeraki.co.ke